Supplementary Material

Effect of alcohol consumption on relapse outcomes among tuberculosis patients: A systematic review and meta-analysis

Running title: Alcohol consumption and tuberculosis relapse

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Supplementary data

Table

Table S1. Search algorithms.
Table S2. Risk of bias assessment of cohort studies included in the meta-analysis by the Newcastle-Ottawa Scale.
Table S3. Description of outcomes in studies included in A meta-analysis.
Table S4. Full-text articles excluded.

Figure

Figure S1. Funnel plot of relapse
Figure S2. Funnel plot of death

Table S1. Search algorithms

Database	No	Step search algorithm	Items found
Pubmed	#1	Search tuberculosis	274,279
	#2	Search "alcohol consumption"	46,884
	#3	Search "alcohol consumption" OR "alcohol drinking"	99,357
	#4	Search "relapse"	669,560
	#5	Search "relapse" OR "reinfection" OR "recurrence" OR "retreatment" OR "failure"	1,556,658
	#6	Search (((tuberculosis) AND (("alcohol consumption)" OR "alcohol drinking") AND ("relapse" OR "reinfection" OR "recurrence" OR "retreatment" OR "failure")	49
Scopus	#1	Search tuberculosis	639,055
	#2	Search "alcohol consumption"	249,986
	#3	Search "alcohol consumption" OR "alcohol drinking"	283,503
	#4	Search "relapse"	508,949
	#5	Search "relapse" OR "reinfection" OR "recurrence" OR "retreatment" OR "failure"	6,120,287
	#6	Search (((tuberculosis) AND (("alcohol consumption)" OR "alcohol drinking") AND ("relapse" OR "reinfection" OR "recurrence" OR "retreatment" OR "failure")	1,454
Cinahl	#1	Search tuberculosis	54,441
	#2	Search "alcohol consumption"	41,249
	#3	Search "alcohol consumption" OR "alcohol drinking"	63,967
	#4	Search "relapse"	62,406
	#5	Search "relapse" OR "reinfection" OR "recurrence" OR "retreatment" OR "failure"	611,944
	#6	Search (((tuberculosis) AND (("alcohol consumption)" OR "alcohol drinking") AND ("relapse" OR "reinfection" OR "recurrence" OR "retreatment" OR "failure")	610

Table S2. Risk of bias assessment of 8 studies included in the meta-analysis by the Newcastle-Ottawa Scale

No.	Author (Year)	Adequacy selection of case-control				Comparability of studies		Outcome assessment			Total NOS score
		Is the case definition adequate?	Representativeness of the cases	Selection of Controls	Definition of Controls	Study control for Age, tb type of tuberculosis	Additional factors; controlled for ≥ 2 variables including Characteristic factors	Ascertainment of exposure	Same method of ascertainment for cases and controls	Non-Response rate	
1	Chen et al. (2019)	✵	✵	✵	✵	✵	✵	✵	✵	-	8/9
2	Cordoba et al. (2020)	✵	✵	✵	✵	✵	✵	✵	✵	-	8/9
3	Kurbatova et al. (2012)	✵	✵	-	-	✵	✵	✵	-	-	5/9
4	Cox et al. (2021)	✵	✵	✵	✵	✵	-	✵	✵	-	8/9
5	Abdelbary et al. (2017)	✵	✵	-	-	✵	✵	✵	-	-	5/9
6	Lampalo et al. (2019)	✵	✵	✵	✵	✵	✵	✵	✵	-	8/9

No.	Author (Year)	Adequacy selection of case-control				Comparability of studies		Outcome assessment			Total NOS score
		Is the case definition adequate ?	Representativeness of the cases	Selection of Controls	Definition of Controls	Study control for Age, tb type of tuberculosis	Additional factors; controlled for ≥ 2 variables including Characteristic factors	Ascertainment of exposure	Same method of ascertainment for cases and controls	Non-Response rate	
7	Peltzer et al. (2012)	✇	✇	-	-	✇	✇	✇	-	-	5/9
8	Bartholomay et al. (2021)	✇	✇	-	-	✇	✇	✇	-	-	5/9

Table S3. Description of outcomes in studies included in meta-analysis

Author (Year)	Outcomes	Definitions
Chen et al., 2019	Failure/ dropout/ death	Recurrent patients in this study met the criteria previously reported in the literature. Essentially, there are two aspects to the criteria for recurrence:(1) completion of the standard anti-TB treatment process and (2) more than two episodes of diagnosed TB (within a minimum time interval of 12 months based on the date of the end of treatment for the first episode). Non-recurrent TB cases are defined as TB patients who test positive for the first time. Relapse, known as endogenous reactivation, is caused by the same strain that caused the first episode of TB.
Cordoba et al., 2020	relapse	The relapse cases were defined as those patients readmitted to the control program due to relapse as defined by the World Health Organization (WHO). A patient previously treated for tuberculosis was declared as cured based on three successive negative smears or negative cultures in the four months after treatment. Controls were defined as those with a first episode of tuberculosis diagnosed in the same year as the case and with no relapse after the same follow-up period.
Kurbatova et al., 2012	Death/ failure/ default	Baseline drug resistance was determined by drug susceptibility tests (DST) in the local laboratory from sputum collected between 60 days prior to 30 days after MDR-TB treatment initiation. Extensively drug-resistant (XDR) TB and SLD groups were defined according WHO guidelines.
Cox et al., 2021	failure / recurrence/ Death	TB treatment outcomes included failure, recurrence or death. Treatment failure was defined as testing positive for TB on sputum microscopy or culture during the last 2 months of treatment. TB recurrence was defined as testing positive for TB on sputum microscopy or culture following successful treatment completion. Death was defined as all-cause mortality within 24 months of treatment initiation.

Author (Year)	Outcomes	Definitions
Abdelbary et al., 2017	treatment failure, death, and drug resistance (all as binary outcomes).	Treatment failure indicated smear-positive persistence at or after 5 months of treatment with first-line anti-TB medications. Death included all causes of death (TB or non-TB related) during the course of TB treatment. Drug resistance was evaluated for five antibiotics (isoniazid, rifampin, pyrazinamide, streptomycin, and ethambutol) using culture methods and coded as present if resistance to any of the five drugs was detected. Multi-drug resistance (MDR) included resistance to isoniazid and rifampicin**.**
Lampalo et al., 2019	Tb relapse	-
Peltzer et al., 2012	Tb relapse	A health care provider who identified a new TB treatment or retreatment patient (within one month on treatment)
Bartholomay et al., 2021	Failure	Failure was defined in the following situations: persistence of smear positive sputum at the end of treatment; cases who had strongly positive smear (++or+ ++) at the beginning of treatment and maintained this situation until the fourth month; initial positive smear followed by negative; and new positive results for two consecutive months, from the fourth month of treatment
	Death	Cases that died from tuberculosis or from other causes during the treatment were considered in the outcome death.

Table S4. Full-text articles excluded

	Author, Year	Reason for exclusion		Author, Year	Reason for exclusion
1	Yen et al., 2017	No outcome of interest	12	Duraisamy et al., 2014	No outcome of interest
2	Shin et al., 2010	No outcome of interest	13	De Albuquerque et al., 2007	No outcome of interest
3	Karl Peltzer et al., 2013	No outcome of interest	14	Ragan et al., 2020	No outcome of interest
4	Peltzer., 2014	Other study design	15	Rehm et al., 2009	Other study design
5	Myers et al., 2018	No outcome of interest	16	Peltzer et al., 2012	Other study design
6	Mulu et al., 2015	No outcome of interest	17	Peltzer and Louw, 2014	Inappropriate population
7	Montiel et al., 2020	Insufficient information	18	Salami and Oluboyo, 2003	No outcome of interest
8	Shin et al., 2013	Other study design	19	De Albuquerque et al., 2001	Other study design
9	Ma, Du, et al., 2019	Other study design	20	dos Santos et al., 2021	No outcome of interest
10	Ma, Che, et al., 2019	No outcome of interest	21	Necho et al., 2021	No outcome of interest
11	Louwagie et al., 2020	Other study design	22	Chenciner et al., 2021	Other study design

Figures

Figure S1. Funnel plot of relapse

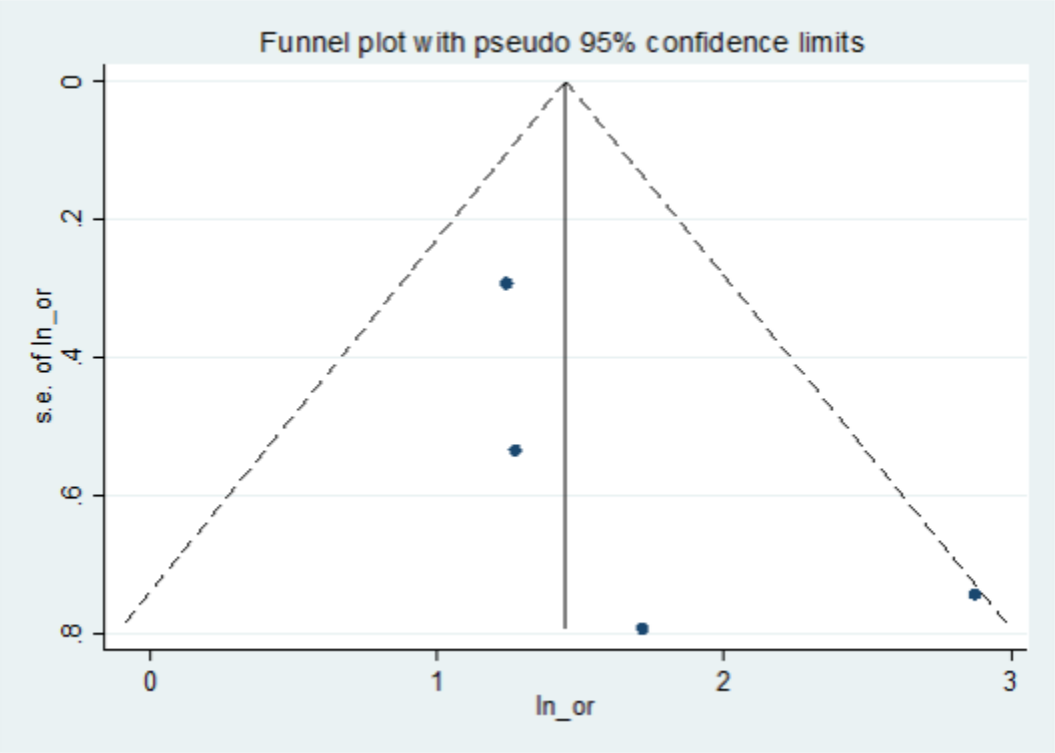
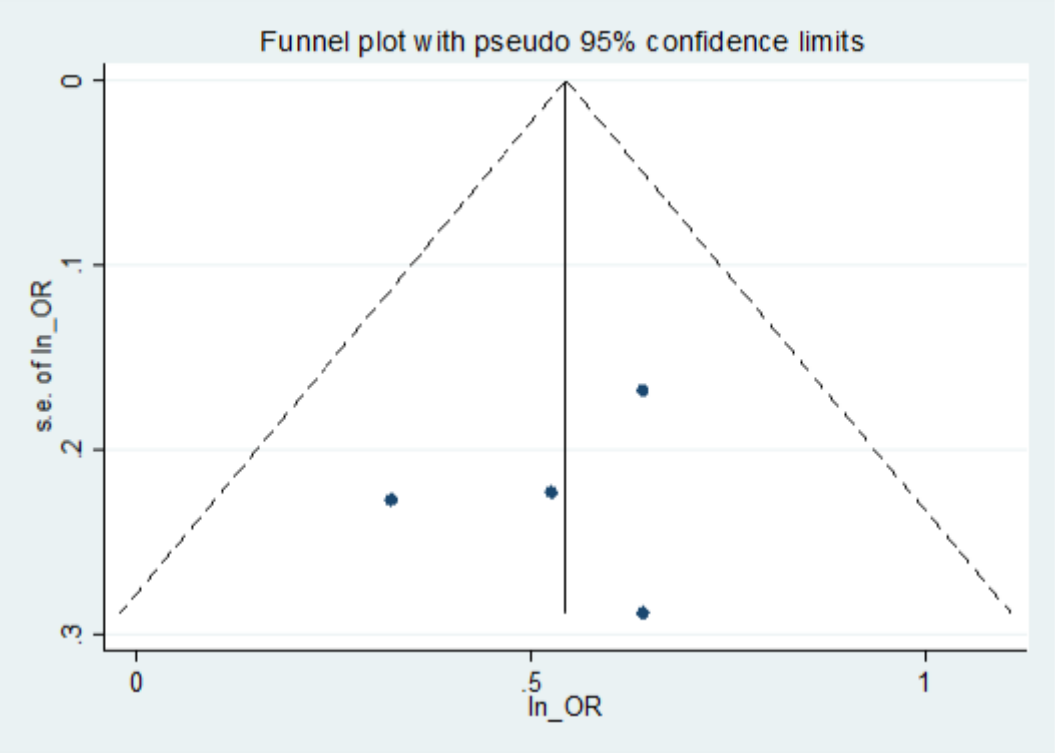


Figure S2. Funnel plot of death



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